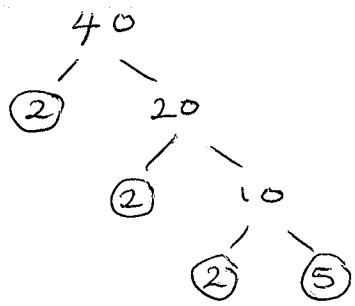
1 Write 40 as a product of its prime factors.



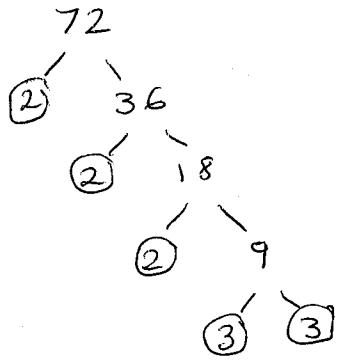
$2 \times 2 \times 2 \times 5$

OR $2^3 \times 5$

$2^3 \times 5$

(Total for question 1 is 2 marks)

2 Write 72 as a product of its prime factors.



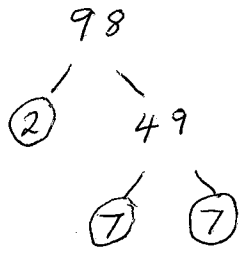
$2 \times 2 \times 2 \times 3 \times 3$

OR $2^3 \times 3^2$

$2^3 \times 3^2$

(Total for question 2 is 2 marks)

3 Write 98 as a product of its prime factors.



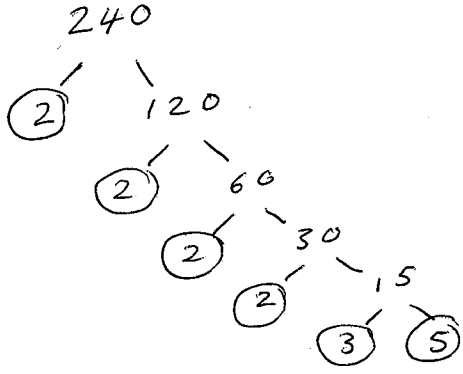
$2 \times 7 \times 7$

OR 2×7^2

2×7^2

(Total for question 3 is 2 marks)

4 Write 240 as a product of its prime factors.



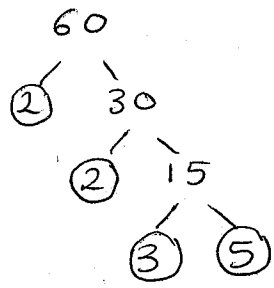
$2 \times 2 \times 2 \times 2 \times 3 \times 5$

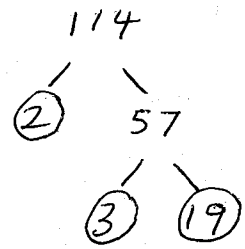
OR $2^4 \times 3 \times 5$

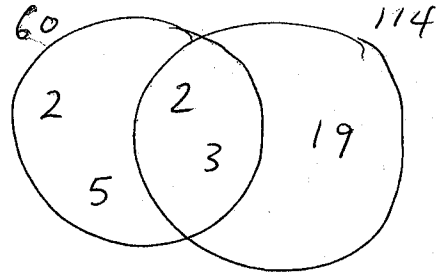
$2^4 \times 3 \times 5$

(Total for question 4 is 2 marks)

5 Find the highest common factor (HCF) of 60 and 114





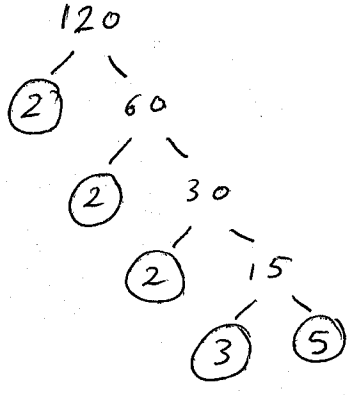


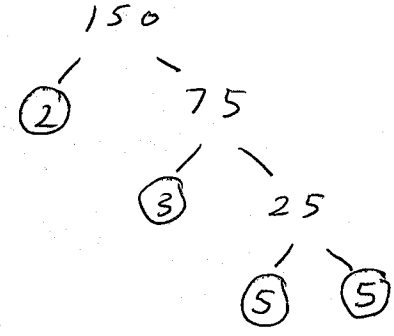
HCF = 2 × 3 = 6

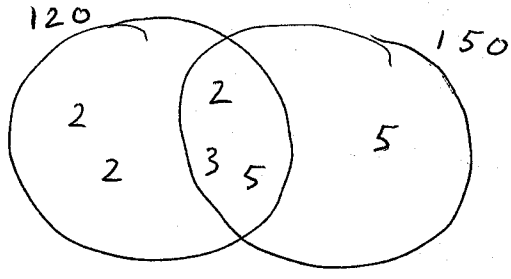
6

(Total for question 5 is 3 marks)

6 Find the lowest common multiple (LCM) of 120 and 150





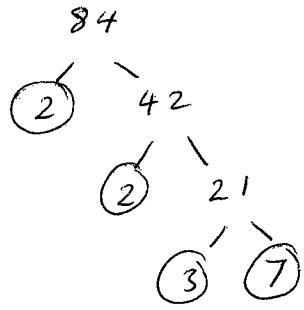


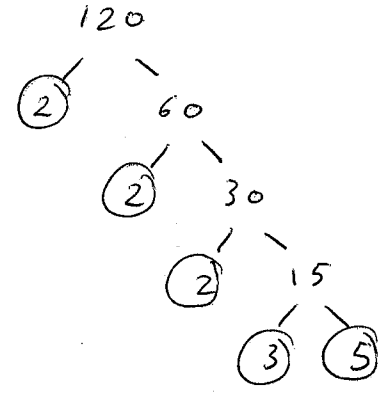
LCM = 120 × 5
= 600

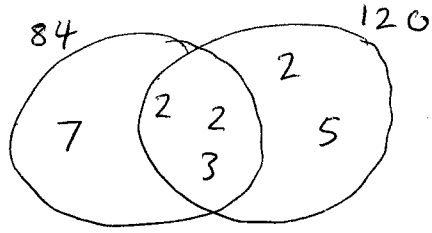
600

(Total for question 6 is 3 marks)

7 Find the highest common factor (HCF) of 84 and 120





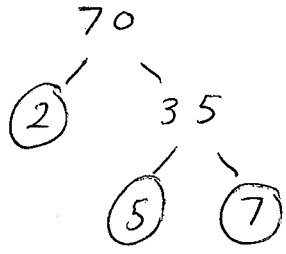


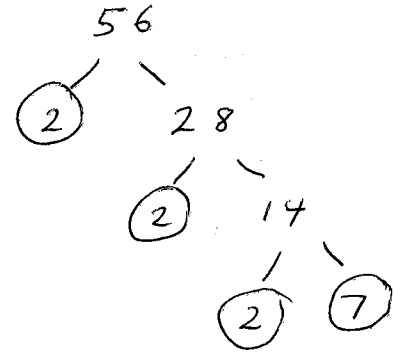
HCF = 2 × 2 × 3
= 12

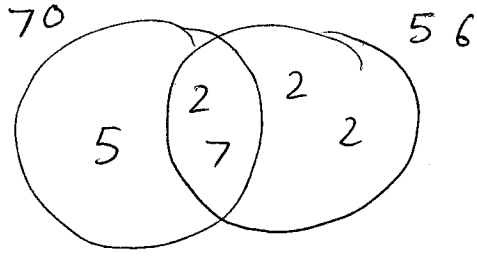
12

(Total for question 7 is 3 marks)

8 Find the lowest common multiple (LCM) of 70 and 56







LCM = 70 × 2 × 2
= 280

280

(Total for question 8 is 3 marks)

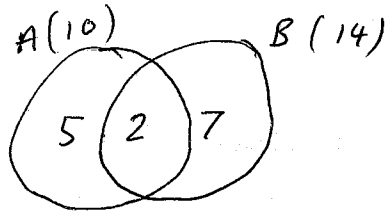
9 Two buses, bus A and bus B, both use the same bus stop.

Bus A runs every 10 minutes.
Bus B runs every 14 minutes.

$10 = 2 \times 5$
$14 = 2 \times 7$

Both buses are at the bus stop at 11 am.

What time will both buses next both be at the bus stop.



LCM $= 14 \times 5$
$= 70$

70 minutes after 11am

12:10 pm

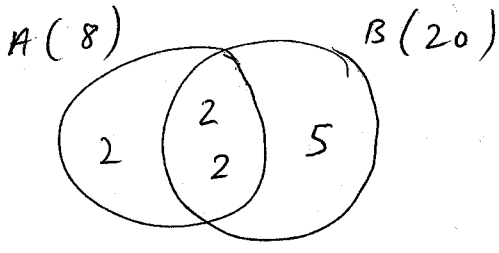
(Total for question 9 is 3 marks)

10 Light **A** flashes every 8 seconds.
Light **B** flashes every 20 seconds.

$8 = 2 \times 2 \times 2$
$20 = 2 \times 2 \times 5$

Both lights flash at the same time.

Work out how long it will take for both lights to flash at the same time again.



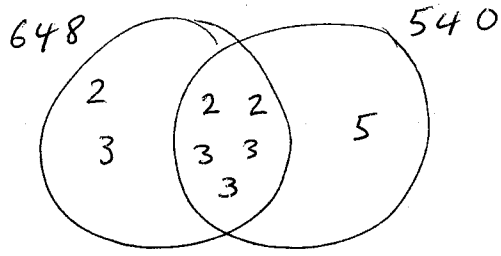
LCM $= 20 \times 2 = 40$

40 seconds

(Total for question 10 is 3 marks)

11 $648 = 2^3 \times 3^4$ $540 = 2^2 \times 3^3 \times 5$

(a) Write down the highest common factor (HCF) of 648 and 540.



$2 \times 2 \times 3 \times 3 \times 3$

4×27

108

(1)

(b) Find the lowest common multiple (LCM) of 648 and 540.

648×5

$\left[\frac{6480}{2} = 3240 \right]$

3240

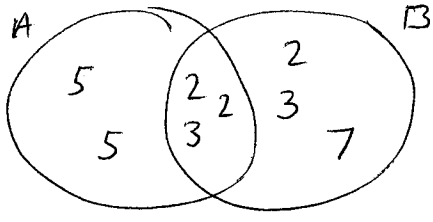
(2)

(Total for question 11 is 3 marks)

---

12 $\mathbf{A} = 2^2 \times 3 \times 5^2$ $\mathbf{B} = 2^3 \times 3^2 \times 7$

(a) Write down the highest common factor (HCF) of **A** and **B**.



$2 \times 2 \times 3$

12

(1)

(b) Find the lowest common multiple (LCM) of **A** and **B**.

$2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7$

$8 \times 9 \times 25 \times 7$

72×175

$$\begin{array}{r} 175 \\ \times 72 \\ \hline 350 \\ 1225 \\ \hline 12600 \end{array}$$

12600

(2)

(Total for question 12 is 3 marks)

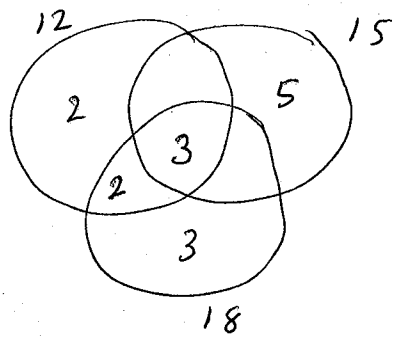
13 Find the lowest common multiple (LCM) of 12, 15 and 18.

~~12 × 2~~

$12 = 2 \times 2 \times 3$

$15 = 3 \times 5$

$18 = 2 \times 3 \times 3$



$LCM = 18 \times 2 \times 5$

$= 18 \times 10$

$= 180$

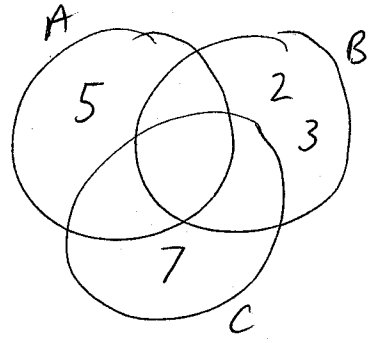
180

(Total for question 13 is 3 marks)

14 Light **A** flashes every 5 seconds. 5
Light **B** flashes every 6 seconds. 2 × 3
Light **C** flashes every 7 seconds. 7

All three lights flash at the same time.

Work out how long it will take for all three lights to flash at the same time again.



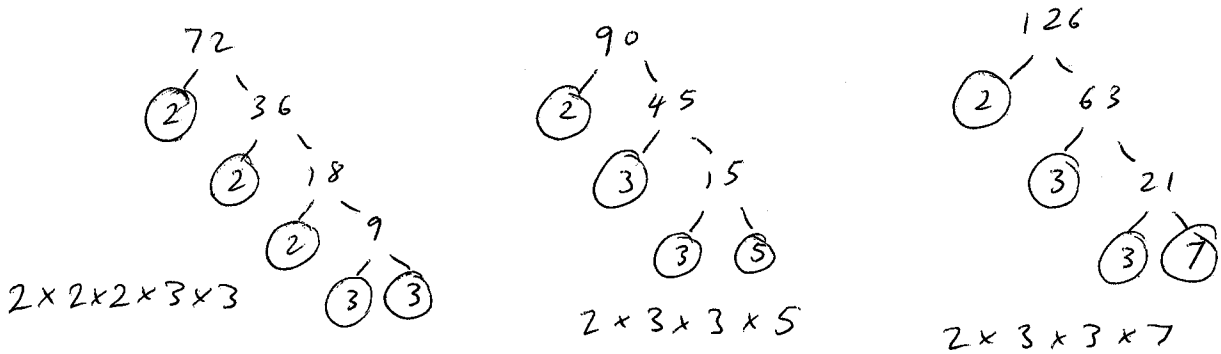
$7 \times 5 \times 6$

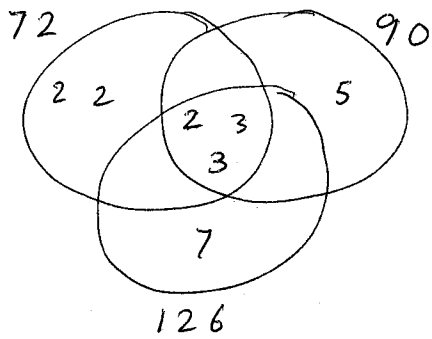
$35 \times 6 = 210$

210 seconds

(Total for question 14 is 3 marks)

15 Find the highest common factor (HCF) of 72, 90 and 126





HCF = 2 × 3 × 3
= 2 × 9
= 18

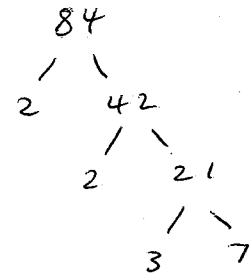
18

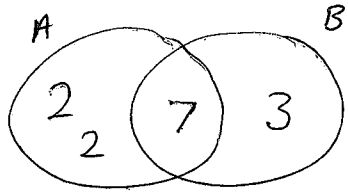
(Total for question 15 is 3 marks)

---

16 Kenny is thinking of two numbers **greater than 10**.
He says: "The highest common factor (HCF) of my two numbers is 7
The lowest common multiple (LCM) of my two numbers is 84"

Write down the two numbers that Kenny is thinking of.





A = 2×2×7
= 28

B = 3×7
= 21

21 and 28

(Total for question 16 is 3 marks)